

The Computational Investigation and Machine Learning Prediction of Fuel Molecular Properties Using Avogadro and Python

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Introduction

As populations rise and countries around the world become increasingly industrialized, global energy needs are growing rapidly. In contemporary society, most of the world's energy is taken from fossil fuels such as petroleum and natural gas. However, a major drawback to this method is that burning such fuels creates and releases carbon dioxide, a gas that traps heat in Earth's atmosphere; this is a large contributing factor in global warming and climate change. The root cause of these emissions is modes of transportation such as cars, trucks, ships, and planes, which all run on petroleum-based fuels (Umenweke and Santillan-Jimenez).

In order to reduce the usage of fossil fuels, scientists and engineers are exploring alternative sources of fuel made from renewable biomass. These are called biofuels, which can be naturally produced from biodegradable materials such as corn, sugarcane, leftover crop materials, cooking oil, and other sources. The most common types of biofuels include ethanol and butanol, which are able to be mixed with gasoline or used as fuel on their own. When compared to harmful fossil fuels, biofuels can significantly lower greenhouse gas emissions, making energy use more eco-friendly ("Biofuels Basics"). In fact, according to the U.S. Energy Information Administration (EIA), new rules have been implemented to require more biofuel

blending and are predicted to push ethanol, biodiesel, and renewable diesel production to record highs in 2026 and beyond. The EIA speculates that fuel ethanol production will continue to increase, resulting in renewable diesel production growing over 20% (Troderman). Because more governments and companies are beginning to search for cleaner energy options, it is imperative to understand how biofuel molecules actually function.

The performance of a fuel is dependent on its molecular structure, specifically: size, molecular weight, polarity, and forces between molecules. These characteristics all affect significant properties such as boiling point, how the fuel needs to be stored, how it is transported, and how well it works as a fuel. Chemical engineers base their studies on these factors, attempting to create improved fuel production systems, separation processes like distillation, and more sustainable energy technologies. The use of software, namely Avogadro, allows scientists and engineers to view, build, and render three-dimensional models of molecules, accelerating research into how differences in structure influence molecular behavior.

Through comparing hydrocarbon fuels such as methane, ethane, propane, and butane with biofuels like ethanol and butanol, this project will explore how molecular structure affects fuel properties and their significance in chemical engineering before expanding the analysis into a machine learning model capable of predicting boiling points from molecular descriptors.

Purpose

The purpose of this project is to explore how molecular structure affects the physical properties of typical hydrocarbon fuels and biofuels using computational modeling, data analysis, and machine learning. By comparing molecular features such as size and functional groups, this

project will demonstrate their impact on the behavior of certain fuels in chemical engineering and how machine learning can predict their boiling points.

Research Question

How does molecular structure affect the physical properties of hydrocarbon fuels and biofuels, and are machine learning models able to accurately predict boiling points based on molecular descriptors and properties? How can this be applied to fuel storage, transportation, and separation processes?

Hypothesis

As the length of the carbon chain increases, the boiling point will increase since bigger molecules undergo stronger intermolecular forces. As well as this, alcohols comprising oxygen atoms will demonstrate increased boiling points compared to similar hydrocarbons due to hydrogen bonding. Moreover, the machine learning models trained on molecular descriptors will be able to reliably predict boiling points, as boiling point is heavily dictated by intermolecular forces and molecular size.

Methods

The six fuels: methane (CH_4), ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}), ethanol ($\text{C}_2\text{H}_5\text{OH}$), and butanol ($\text{C}_4\text{H}_9\text{OH}$) were selected and analyzed throughout the experiment. These molecules were specifically chosen because they are common hydrocarbon fuels and biofuels used in transportation and energy.

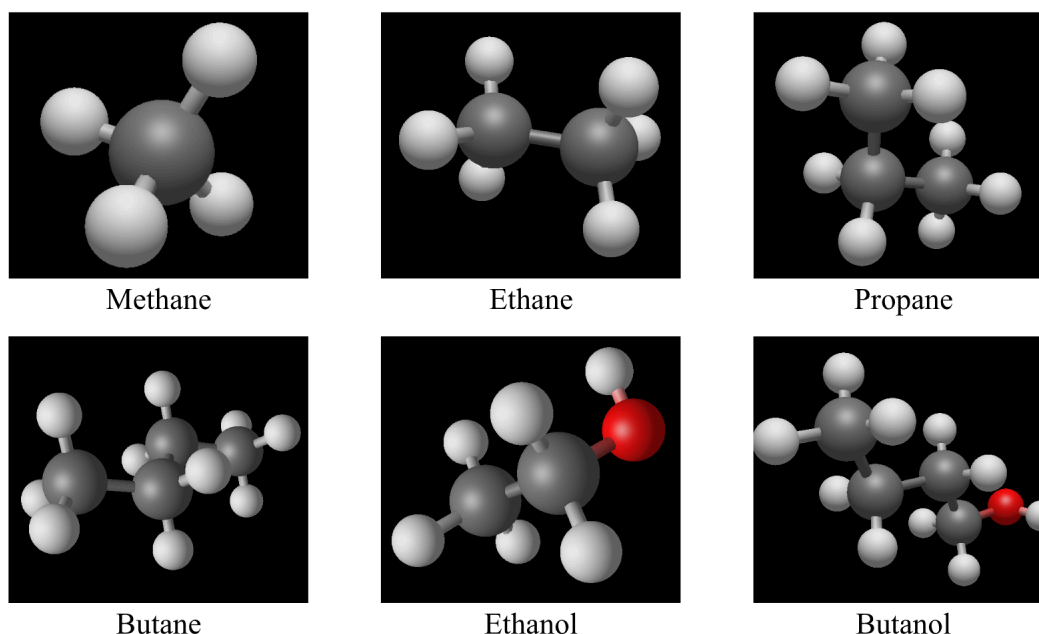


Figure 1. Optimized molecular models of methane, ethane, propane, butane, ethanol, and butanol created using Avogadro

The molecular structures were created from Avogadro's modeling tools, where each molecule's geometry was optimized to generate a stable three-dimensional structure. Screenshots of each molecule were taken for comparison.

The bond lengths and angles were measured using Avogadro's measurement tools, where carbon-carbon (C-C), carbon-hydrogen (C-H), carbon-oxygen (C-O), and oxygen-hydrogen (O-H) bonds were evaluated. Bond angles such as H-C-H, C-C-C, and C-C-O were also measured.

Following this initial investigation, the study was then expanded to include 20 fuel molecules covering six chemical families: alkanes, alcohols, ethers, esters, ketones, and aromatic hydrocarbons. Each molecule was created and measured using the same procedure in Avogadro.

Additionally, physical and chemical properties for each molecule, such as molecular weight and rotatable bonds, were acquired from PubChem and incorporated into the dataset.

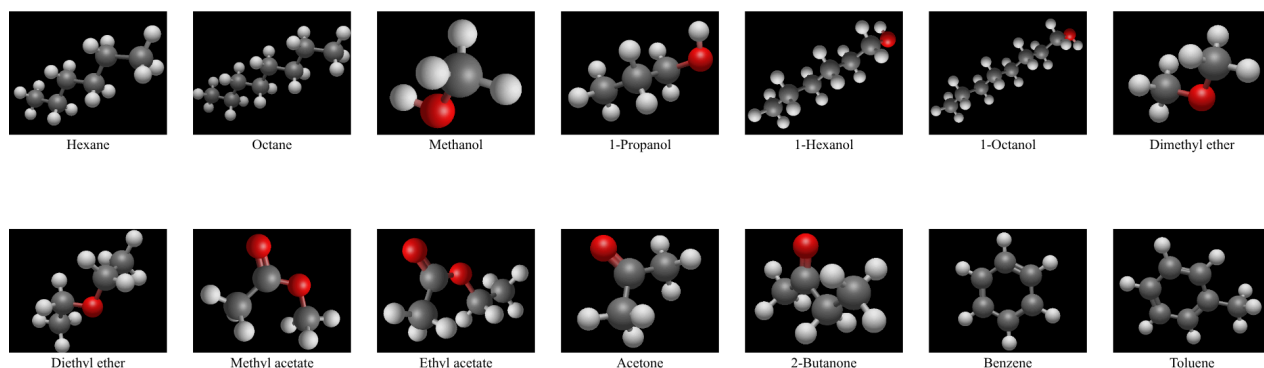


Figure 2. Optimized molecular models of the additional fuel molecules created using Avogadro

The dataset was then expanded to 100 fuel molecules, introducing aldehydes and carboxylic acids to broaden the dataset for machine learning analysis. The larger dataset omitted bond lengths and angles because manually measuring these values for all 100 molecules would have been impractical. Despite this, the other molecular descriptors were retained and used to train the machine learning models.

Categorical variables were converted into integers using one-hot encoding, and the dataset was divided into training and testing sets using an 80:20 ratio. Four regression models were evaluated: Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression. Their performance was measured using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). Feature importance was calculated using the Random Forest model to determine which descriptors had the largest influence in predicting the boiling point.

All of the datasets were created using Google Sheets and analyzed in Python using Google Colab. Claude AI code generation was used, but all of the code was tested, reviewed, and revised to verify functionality. This was done through practical understanding of the Pandas library for data organization, Matplotlib for data visualization, and Scikit-learn for evaluation of the machine learning models.

All of the source code, datasets, models, and workflow can be accessed in the project's GitHub repository: <https://github.com/matthewdengg/computational-fuel-analysis>

Results

The six fuels were generated and analyzed in Avogadro, specifically the four hydrocarbon fuels and two biofuels.

Molecule	Formula	Contains Oxygen?	Functional Group
Methane	CH ₄	No	Alkane
Ethane	C ₂ H ₆	No	Alkane
Propane	C ₃ H ₈	No	Alkane
Butane	C ₄ H ₁₀	No	Alkane
Ethanol	C ₂ H ₅ OH	Yes	Alcohol (-OH)
Butanol	C ₄ H ₉ OH	Yes	Alcohol (-OH)

Table 1. Molecular properties and characteristics of the analyzed fuel molecules

Following the examination, methane, ethane, propane, and butane were found to be all alkanes made up of only carbon and hydrogen atoms. Ethanol and butanol, on the other hand, contained oxygen in the form of a hydroxyl (-OH) functional group.

The bond length was generally consistent for all molecules.

Molecule	Measured Bond	Length (Å)
Methane	C-H	1.109
Ethane	C-C, C-H	1.517, 1.110
Propane	C-C	1.52
Butane	C-C	1.52
Ethanol	C-C, C-O, O-H	1.519, 1.110, 1.398
Butonal	C-C, C-O, O-H	1.524, 1.111, 1.398

Table 2. Bond length measurements across the analyzed molecules

C-C bonds ranged from 1.517 Å to 1.524 Å, while C-H bonds ranged from 1.109 Å to 1.111 Å. On top of this, the C-O and O-H bonds remained approximately 1.398 Å in both alcohol molecules. These findings demonstrate that the bond lengths between each atom had little to no variation despite differences in molecular size and composition.

Likewise, the bond angles showed little variation.

Molecule	Measured Angle	Bond Angle (°)
Methane	H-C-H	109.471
Ethane	H-C-H	109.119
Propane	C-C-C	111.324
Ethanol	C-C-O	110.499
Butonal	C-C-O	110.327

Table 3. Bond angle measurements across the analyzed molecules

The measured bond angles for each molecule ranged from around 109° to 111°, staying relatively close to the ideal tetrahedral angle of 109.5°. This reveals that the molecular geometry was similar between each molecule despite differences in functional groups and size.

However, the physical property data displayed a much more noticeable variation than the geometric measurements.

Molecule	Weight (g/mol)	Boiling Point (°C)
Methane	16.04	-161.5
Ethane	30.07	-88.6
Propane	44.1	-42.1
Butane	58.12	-0.5
Ethanol	46.07	78.4
Butanal	74.12	117.1

Table 4. Physical properties of the analyzed fuel molecules

The molecular weight gradually increased from methane (16.04 g/mol) to butanol (74.12 g/mol). The boiling point also increased simultaneously with molecular size. Boiling points rose from -161.5°C for methane to -0.5°C for butane, with the alcohols having significantly higher boiling points than hydrocarbons with similar carbon chain lengths. For example, ethanol had a boiling point of 78.4°C, compared to 117.1°C for butanol.

The first graph depicted an overall positive relationship between various properties and boiling point.

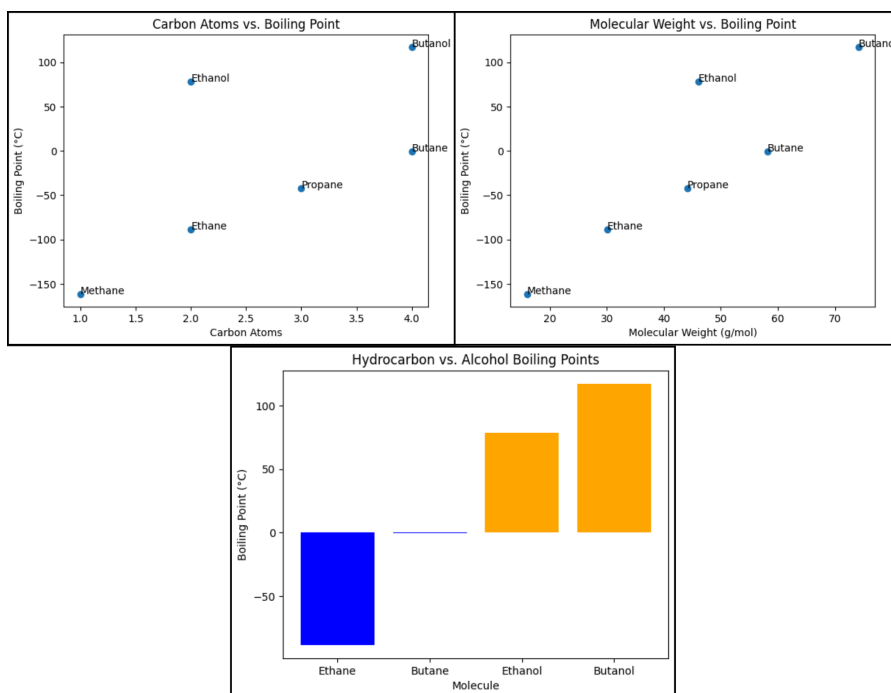


Figure 3-5. Relationship between various properties and boiling point

As shown, boiling point increased with molecular size and was more noticeable for alcohols than for hydrocarbons of similar size.

After discovering these trends, the dataset was then expanded to 20 molecules across six chemical families for further investigation: alkanes, alcohols, ethers, esters, ketones, and aromatic hydrocarbons.

Variable	Value
Number of molecules	20
Chemical families	6
Weight range (g/mol)	16.04-130.23
Boiling point range (°C)	-161.5-195.0
Carbon chain range	1-8

C-C range (Å)	1.390-1.524
C-H range (Å)	1.082-1.111
C-O range (Å)	1.373-1.406
O-H range (Å)	0.991-0.992
C=O range (Å)	1.217-1.219
Bond angle range (°)	105.954-120.280
Rotatable bond range	0-6

Table 5. Summary of the 20 analyzed fuel molecule dataset

The 14 additional molecules continue following the same trends observed in the six-molecule investigation. The bond lengths and angles stayed consistent across each family, while boiling point fluctuated with molecular weight, carbon chain length, and functional groups. Specifically, alcohols capable of hydrogen-bonding consistently had higher boiling points than ethers and hydrocarbons of similar size. The complete dataset can be found here:

<https://docs.google.com/spreadsheets/d/1OK9FG9tsYg8NmCBICVp4U5GAjZ70IBB7fujPPoeRwic/edit?usp=sharing>

Property	Mean	Median	Std Dev	Min	Max	Range
Weight (g/mol)	68.81	73.1	28.96	16.04	130.23	114.19
Boiling Point (°C)	54.06	72.9	83.68	-161.50	195	356.5
Carbon Chain Length	3.75	3	2.31	1	8	7
Bond Angle (°)	112.87	110.98	4.57	105.95	120.28	14.33

Rotatable Bonds	1.4	1	1.82	0	6	6
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Table 6. Statistical summary of the 20-molecule dataset

The dataset showed a large variation in molecular weight and boiling point, while bond angles displayed smaller changes.

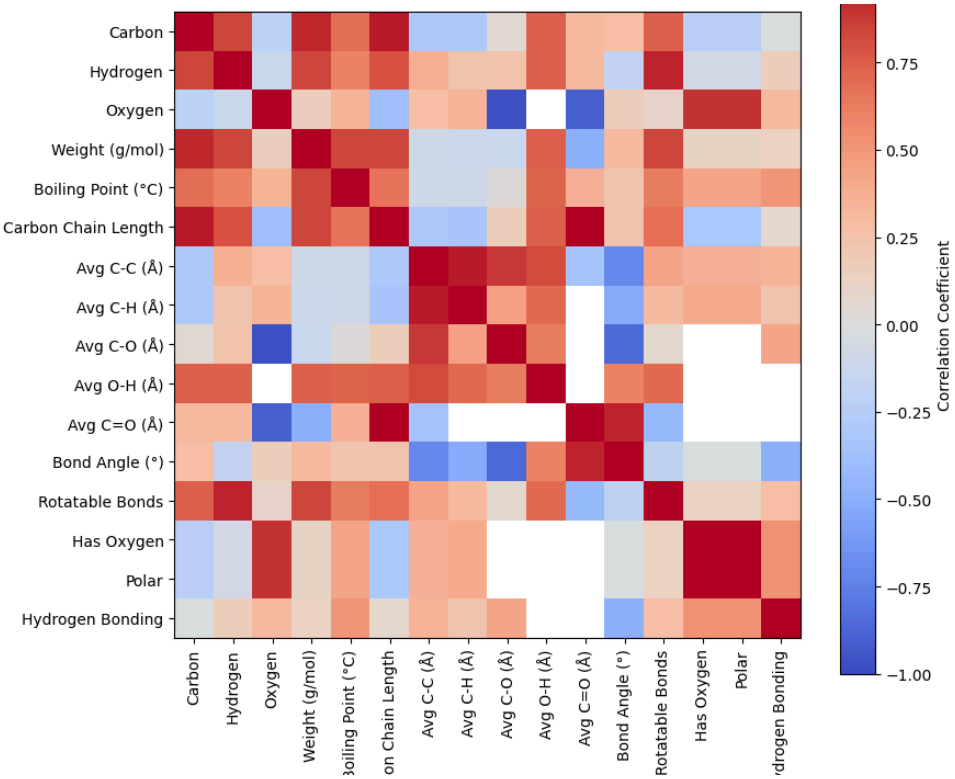
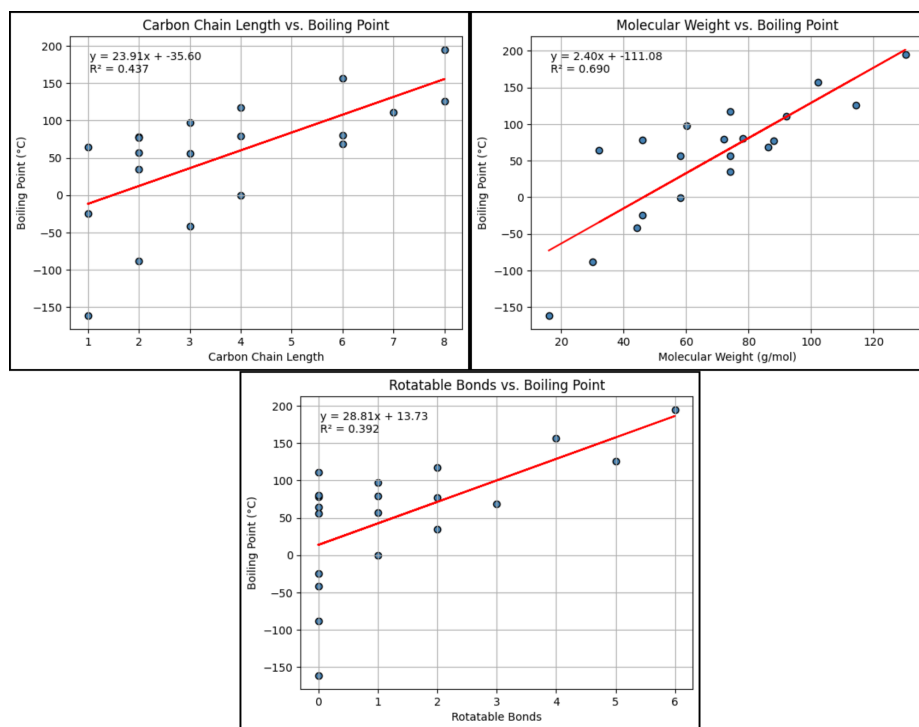


Figure 6. Correlation matrix based on molecular descriptors

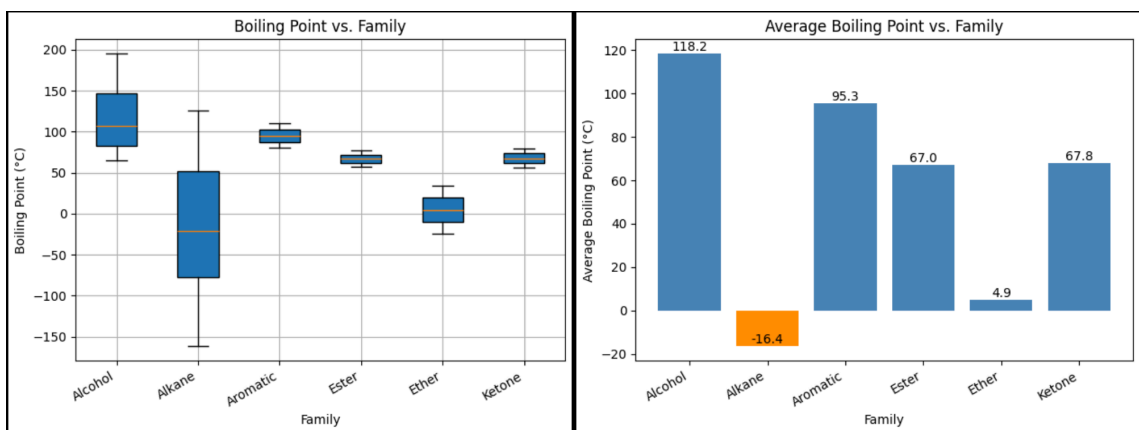
Based on Figure 6, it is evident that molecular weight had the strongest correlation with boiling point among the descriptors, with a correlation coefficient of 0.831. The carbon presence, carbon chain length, and rotatable bonds also displayed a strong correlation, with values between 0.626 and 0.695. However, the average C-C, C-H, and C-O bond lengths demonstrated small

correlations with boiling point, highlighting that bond lengths are not reliable predictors of boiling point compared to the broader molecular properties.



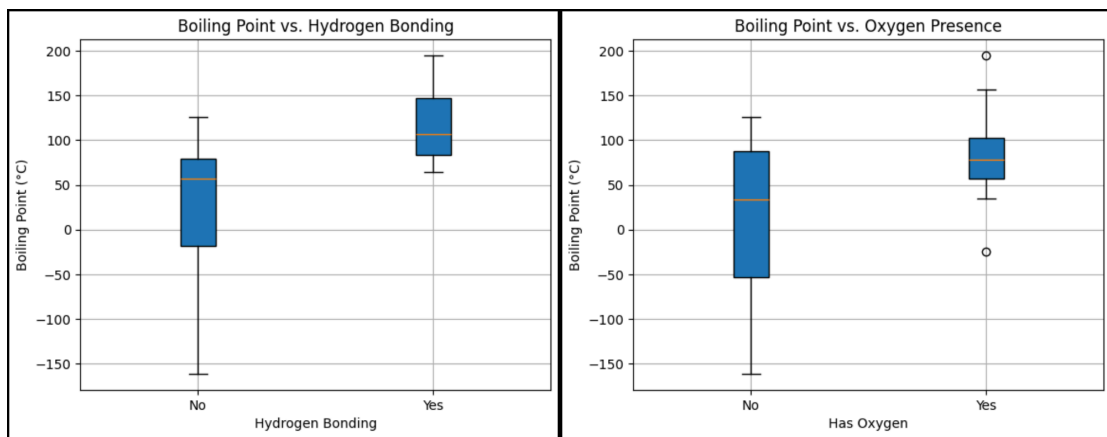
Figures 7-9. Relationship between several molecular descriptors and boiling point

The generated scatterplots display a positive correlation between molecular properties and boiling point. With that being said, molecular weight demonstrated the strongest relationship, consistent with its high correlation coefficient. Ultimately, each plot was consistent with the trends seen in the correlation matrix.



Figures 10-11. Comparison between boiling point and chemical families

The chemical families were also observed to have distinct physical properties. The greatest variations in boiling point compared to the rest of the families were in alcohols and alkanes. Alcohols had higher average boiling points than hydrocarbons with similar molecular sizes since their hydroxyl groups allowed hydrogen bonding. These findings show that boiling points differed among each chemical family, even for molecules of similar size.



Figures 12-13. Comparison between molecular properties and boiling point

Molecules that were capable of hydrogen bonding generally had a higher boiling point than those that could not. Correspondingly, molecules with oxygen commonly had higher boiling points than those without oxygen as well.

Building upon these trends, the dataset was then expanded to 100 molecules to train machine learning models that predict boiling points using molecular descriptors.

Chemical Family	Number of Molecules
Alkane	10
Alcohol	15
Ether	14
Ester	15
Ketone	14
Aromatic	13
Aldehyde	10
Carboxylic Acid	9

Table 7. 100-molecule dataset composition

The dataset retained the same molecular descriptors as the previous dataset, except for bond angle and length, and added heavy atom count for another measure of molecular size.

Property	Mean	Median	Std Dev	Min	Max	Range
Weight (g/mol)	96.53	100.16	31.19	16.04	186.33	170.29
Boiling Point (°C)	111.66	118.7	67.18	-161.50	254	415.5
Carbon	5.68	6	2.46	1	12	11

Chain Length						
Heavy Atom Count	6.61	7	2.28	1	13	12
Rotatable Bonds	2.55	2	2.19	0	10	10

Table 8. Statistical summary of 100-molecule dataset

The finalized dataset displayed a wider range of molecular and physical properties than the previous dataset. The molecular weight range increased from 16.04 g/mol for methane to 186.33 g/mol for dihexyl ether, while boiling point increased from -161.5°C for methane to 254.0°C for nonanoic acid. The carbon chain length and rotatable bonds followed the same pattern, with chain lengths ranging from 1 to 12 carbon atoms and bond counts ranging from 0 to 10. With this in mind, the newly added property, heavy atom count, was found to be within the range of 1 to 13 atoms.

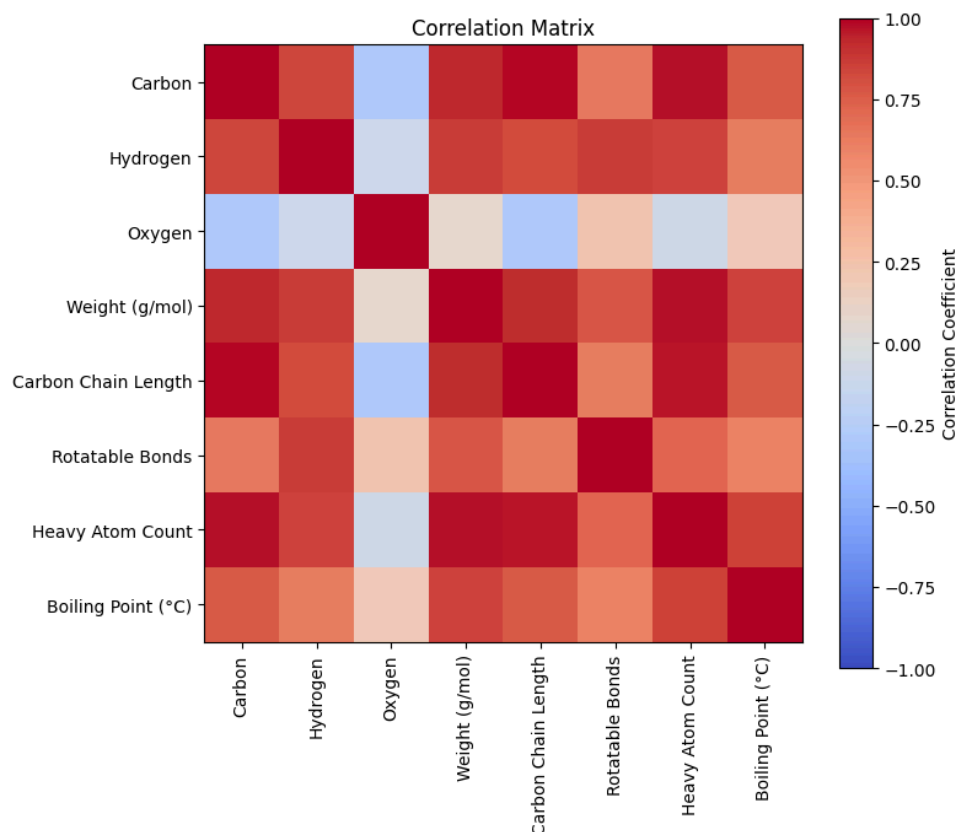
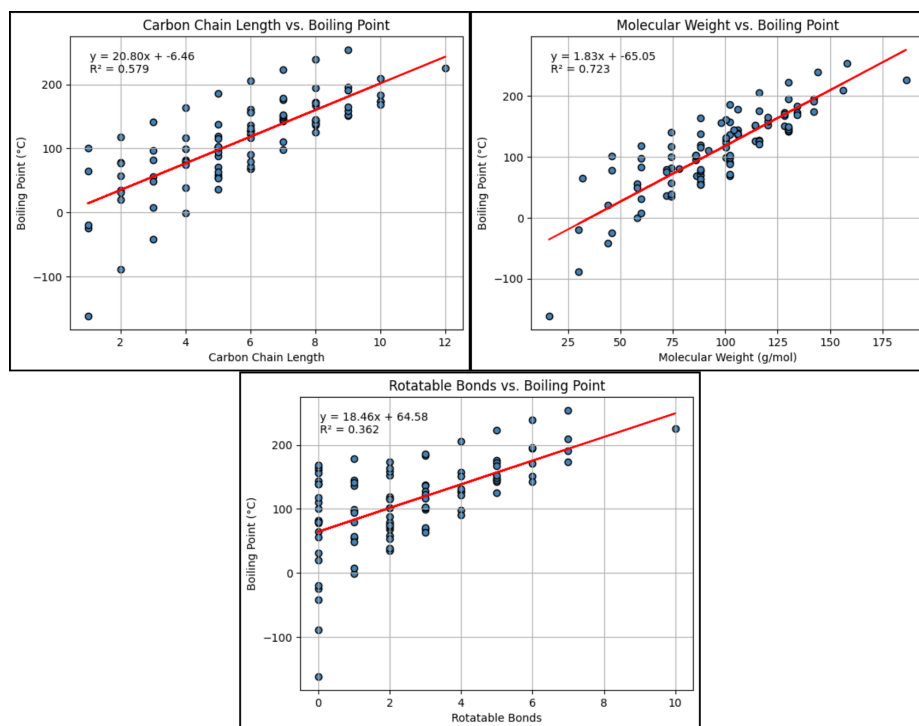


Figure 14. Correlation matrix based on molecular descriptors for the 100-molecule dataset

According to Figure 14, heavy atom count held the strongest correlation with boiling point, with a correlation coefficient of 0.855, with molecular weight following closely at 0.850. Compared to the 20-molecule dataset, carbon chain length and carbon presence correlation values increased to 0.760 and 0.759, respectively. Despite this, rotatable bonds were the only descriptor that showed a decrease in correlation, reducing to 0.602. These findings show that properties related to molecular size tend to have the strongest relationships with boiling point.



Figures 15-17. Relationship between several molecular descriptors and boiling point for the 100-molecule dataset

Each figure demonstrates a positive correlation between each molecular property and boiling point. As with the 20-molecule dataset, Figure 16 showed the strongest relationship, consistent with its high correlation coefficient, supporting the correlation matrix.

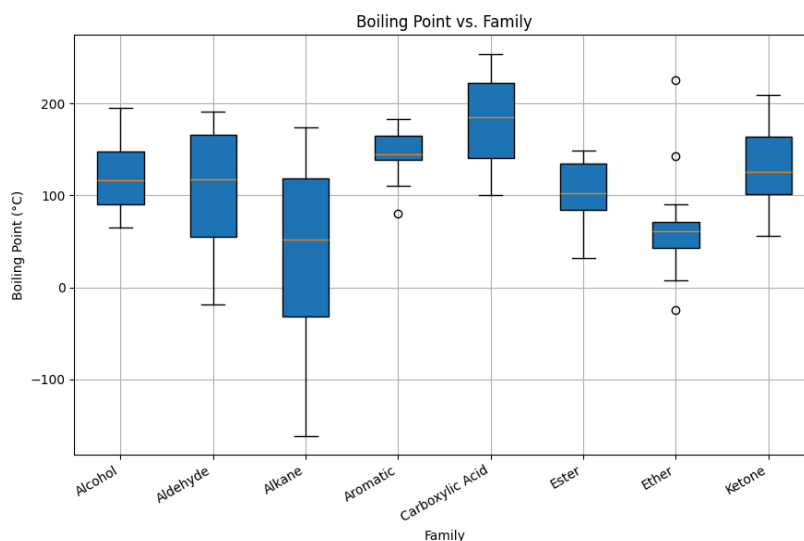


Figure 18. Comparison between boiling point and chemical family for the 100-molecule dataset

As shown in Figure 18, the greatest variations in boiling point were in alkanes and aldehydes. Carboxylic acids are also shown to have higher average boiling points than the other chemical families.

After reanalyzing the relationships between molecular descriptors and boiling point, four machine learning regression models were trained on the 100-molecule dataset to determine whether boiling points could be predicted from molecular properties.

Model	R ²	MAE	RMSE
Linear Regression	0.8646	14.7483	27.668
Decision Tree	0.8063	21.1725	33.093
Random Forest	0.8247	17.9481	31.475
Gradient Boosting	0.9334	11.9931	19.4138

Table 9. Machine learning model performance

Each analyzed model was observed to generate varying degrees of performance. Gradient Boosting delivered the best results, yielding the highest R^2 value of 0.9334 alongside the lowest MAE and RMSE values of 11.99°C and 19.41°C. Linear Regression produced the second largest R^2 value of 0.8646, while Random Forest and Decision Tree produced lower values of 0.8247 and 0.8063, respectively. The high R^2 value of Gradient Boosting shows that the model was able to explain most of the variation in boiling point. The MAE indicates that its predictions deviated from the actual boiling points by approximately 12°C on average, whereas the larger RMSE suggests that some predictions contained greater errors than others.

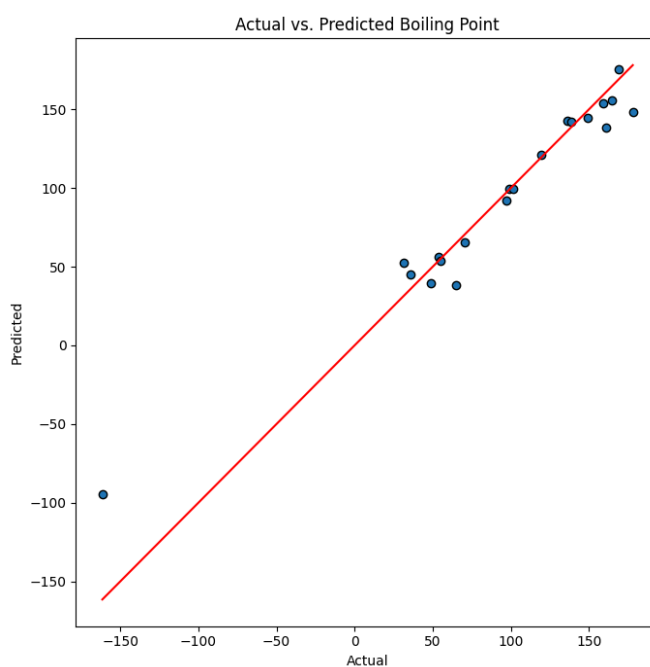


Figure 19. Comparison between empirical and predicted boiling point

Figure 19 compares the actual boiling points of the molecules with the values predicted by the Gradient Boosting model. Most of the plotted points were near the reference line, showing

that the predicted values were typically close to the actual boiling points. Altogether, the graph is consistent with the model's high R^2 value of 0.9334.

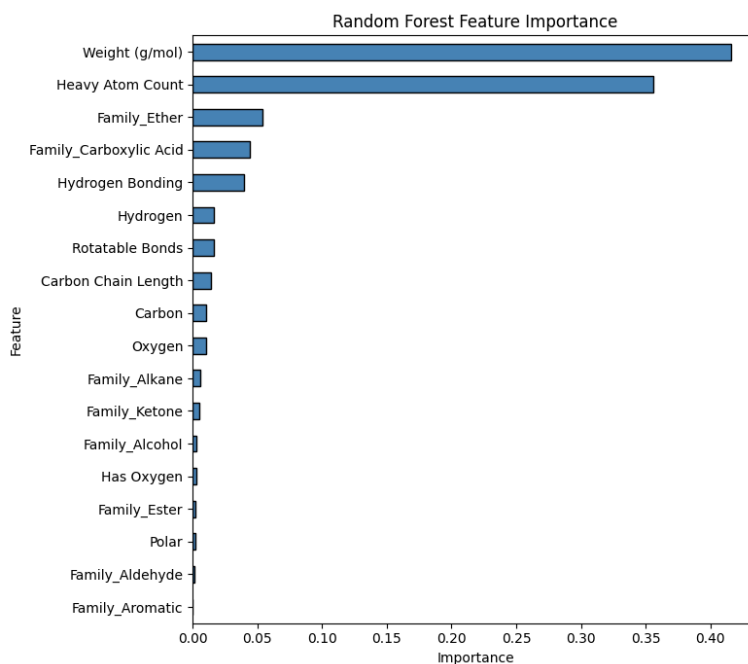


Figure 20. Random forest feature importance graph based on molecular descriptors

Based on the plot, it is evident that molecular weight was the most impactful descriptor, carrying a feature importance of 0.416, followed by heavy atom count at 0.356. Several chemical families and hydrogen bonding had a notable contribution to the model as well.

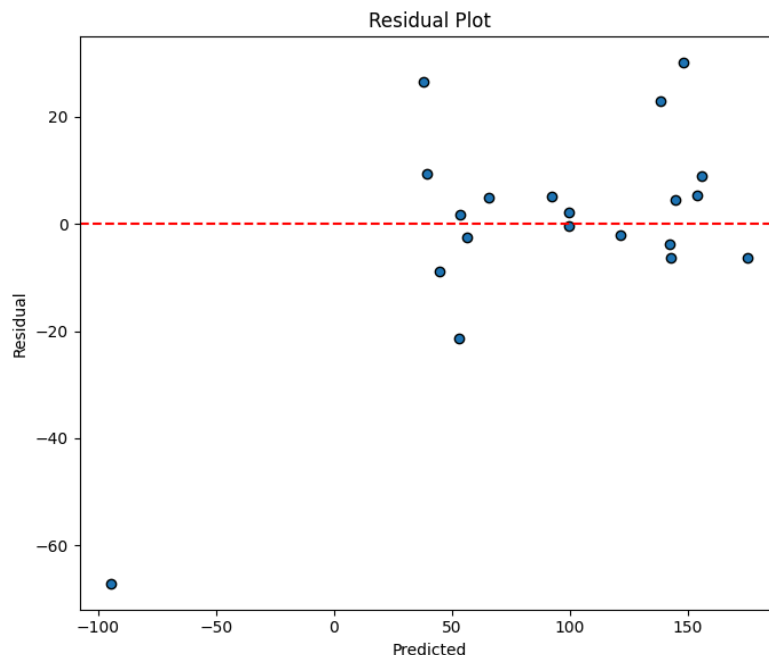


Figure 21. Residual plot

As displayed in Figure 21, most residuals were plotted around zero. This demonstrates that the model didn't consistently overpredict or underpredict boiling points across the dataset. Despite this, there were several larger residuals that both heavily underpredicted and overpredicted the boiling points, showing that the model had difficulties in predicting certain molecules. For clarity, individual predictions from the Gradient Boosting model can be found here:

https://docs.google.com/spreadsheets/d/1mjKIa2hRsm0YFfuwD3tmFKSCONYh_5JxmSdAcJEgIfw/edit?usp=sharing

Discussion

The results support both hypotheses. As the carbon chain length increased from methane to butane, the molecular weight and boiling point increased accordingly since larger molecules

experience stronger intermolecular forces. Stronger intermolecular attractions require more energy to overcome during boiling, leading to a consistently higher boiling point. Alcohols were found to have higher boiling points than hydrocarbons of similar size due to their ability to form hydrogen bonds. On top of this, the machine learning study showed that boiling point could be accurately predicted based on molecular descriptors, with the Gradient Boosting model producing the most reliable prediction ($R^2 = 0.9334$).

As shown in Tables 2 and 3, the bond lengths and angles remained relatively constant despite different molecules being measured. C-C bond lengths stayed around 1.52 Å, C-H bonds averaged close to 1.11 Å, and most bond angles were close to the ideal tetrahedral angle of 109.5°. This displays that molecular geometry was not the primary factor in dictating boiling point, and that structural properties such as molecular size and functional groups are much more impactful.

With that being said, the presence of oxygen within the molecules has demonstrated a much more profound impact on fuel behavior. Ethanol and butanol, which both contain hydroxyl functional groups, can form hydrogen bonds between molecules. Hydrogen bonding is significantly stronger than the London dispersion forces commonly found in hydrocarbons. Consequently, ethanol and butanol exhibited substantially higher boiling points than ethane and butane despite having similar carbon chain lengths. This displays how intermolecular forces are a much more prevalent factor in determining boiling point than small variations in molecular geometry.

Increasing the size of the dataset from 6 to 20 molecules and ultimately to 100 molecules made the observed trends and relationships much more apparent. Among the analyzed datasets, molecular weight, heavy atom count, carbon chain length, and rotatable bonds all produced

positive relationships with boiling point. The correlation analysis also displayed molecular weight and heavy atom count as the strongest predictors, while bond lengths exhibited very weak correlations. These results suggest that molecular structure can provide more information about boiling point than bond measurements.

In fact, when incorporating machine learning models, Gradient Boosting was found to have the highest prediction accuracy with an R^2 value of 0.9334 while simultaneously producing the lowest MAE and RMSE values. Molecular weight and heavy atom count were also found to be the most influential predictors in the feature importance analysis, further supporting the relationships found throughout the study. These findings confirm that machine learning can effectively estimate boiling point using molecular descriptors alone.

This knowledge is crucial for chemical engineers, as they must understand how molecular structures influence physical properties when designing fuel storage systems, transportation methods, and industrial separation processes such as distillation. As well as this, machine learning models can allow engineers to estimate fuel properties before testing, preventing errors, reducing development time and costs. For example, butane's low boiling point means that it must be stored under moderate pressure in sealed canisters, the most common method being in portable camping stoves and household lighters. In contrast, butanol's high boiling point makes it much easier to handle and distribute as a biofuel. Similarly, the drastic difference in boiling points between ethanol and butane allows them to be separated efficiently using fractional distillation, a physical process used in fuel refineries to separate a complex mixture of liquids into individual components based on their boiling points (Darcel).

With these findings in mind, one limitation of this investigation is that only 100 molecules were used to train the machine learning models, with only a small number of

molecular descriptors being analyzed. The addition of other descriptors, such as molecular surface area, ring structure, or dipole moment, could improve prediction accuracy. Moreover, this study focused solely on boiling point; further exploration could include other fuel properties such as density, viscosity, and flash point.

Overall, the inquiry demonstrates that molecular geometry strongly influences the physical properties of fuels. Though the bond lengths and angles remained relatively constant, it was ultimately molecular size, functional groups, and intermolecular forces that drove the boiling point. Expanding the study from 6 to 100 molecules reinforced these relationships, allowing accurate machine learning predictions of boiling points. As a whole, combining computational chemistry with machine learning creates a reliable method for understanding fuel behavior, opening up new possibilities for future chemical engineering research.

Conclusion

This study evaluated how molecular structure affected the physical properties of common hydrocarbon fuels and biofuels using Avogadro molecular modeling, Python data analysis, and machine learning models. Initially starting with six fuel molecules in Avogadro, the investigation gradually expanded the dataset to 20 molecules and eventually 100 molecules. In doing so, broader relationships between molecular descriptors and boiling point were able to be discovered and analyzed. The results directly supported the hypothesis that increasing carbon chain lengths leads to higher boiling points since larger molecules experience stronger intermolecular forces. Moreover, alcohols containing hydroxyl groups regularly had significantly higher boiling points than hydrocarbons with similar carbon chain lengths due to hydrogen bonding.

Despite bond lengths and bond angles being relatively similar among the molecules, molecular composition had a much stronger impact on physical properties. Correlation analysis displayed that molecular weight, heavy atom count, carbon chain length, and rotatable bonds were the most reliable predictors of boiling point, with bond lengths having little contribution. This shows how molecular structure is a much stronger indicator of boiling point than geometrical measurements.

Gradient Boosting outperformed the other four regression models as well, reaching an R^2 value of 0.9334. This proves that boiling point can be predicted pretty accurately using just molecular descriptors. The feature importance analysis also pointed to molecular weight and heavy atom count as being the most influential properties.

All in all, this project combines computational chemistry, data analysis, and machine learning to investigate molecular behavior. Computational tools such as Avogadro and Python make it possible to visualize molecular structures, explore relationships between molecular and physical properties, and accurately predict important physical properties. As a whole, understanding these connections is vital for chemical engineers working to improve fuel storage, transportation, separation processes, and the development of increasingly sustainable energy technologies.

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